

GPU Requirement & Selection

Infrastructure sizing for the voice AI PoC stack

1. Memory Requirement

All three models run simultaneously on the same machine.

Component	Model	Memory
Speech-to-Text (STT)	Qwen3-ASR-1.7B	~6–8 GB
Intent LLM	Sahabat-AI 9B (int8)	~10 GB
Dialogue LLM	SEA-LION v4 / Qwen3-32B (FP8-quantized)	~32 GB
Buffer	Concurrent requests, KV cache	~10–15 GB
Total required		~60–65 GB

Implication: 24GB-class cards cannot hold the full stack. We need either one large GPU or two mid-size GPUs.

2. GPU Options

Options are ranked by fit and economy.

GPU	VRAM	Fits the stack?	Provider & rate	Notes
1x A100 80GB (Recommended)	80 GB	Yes. All 3 models with headroom	RunPod: ~\$1.64/hr Lambda Labs: ~\$1.79/hr	Best overall pick. Single machine, simplest setup, real headroom.
2x L40S 48GB	96 GB total	Yes. Models split across two GPUs	RunPod: ~\$0.86/hr each (~\$1.72/hr total)	Similar cost to one A100, but adds complexity from splitting models. Only worth it if the A100 is unavailable.
1x A6000 48GB	48 GB	Tight. STT and intent model fit, but the 32B dialogue model does not fit well	RunPod: ~\$0.49/hr	Cheapest, but too small for the full stack. Would force a smaller dialogue model and change the plan.
1x H100 80GB	80 GB	Yes. More headroom and speed	RunPod: ~\$2.69/hr Lambda: ~\$2.99/hr	Overkill at PoC scale. Meaningfully pricier for speed we don't need yet.

3. Recommendation

Selected GPU	1x NVIDIA A100 80GB
Provider	RunPod
Rate	~\$1.64/hr
Why	Best balance of capacity, simplicity, and cost for the PoC

Note: Rates are approximate and should be re-verified before purchase, as cloud GPU pricing changes frequently.